

Chanyoung Park

CONTACT INFORMATION

Associate Professor
[Data Science & Artificial Intelligence Lab \(DSAIL\)](#)
 Dept. of Industrial & Systems Engineering, KAIST
 The Kim Jaechul Graduate School of AI, KAIST
[Graduate School of Data Science, KAIST](#)
 Room #4104, E2-2 (Industrial & Management Building)

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 Homepage: <https://dsail.kaist.ac.kr>

[Google Scholar](#)

CURRENT RESEARCH

- **AI for Scientific Discovery**
 - Materials, Chemistry
 - Biology, Engineering, EDA
- **Agentic AI**
- **LLM / Multimodal Data Mining**
 - Recommender System, Graph Neural Networks
 - LLM Safety, Model Efficiency, Video Language Model
 - Scene Graph Generation, Anomaly Detection, Time Series Analysis
- **Data-Centric AI**
 - Model Robustness, Adversarial Attack/Defense
 - Imbalanced Learning, Few-shot Learning, Explainability

POSITIONS

- | | |
|--|---------------------|
| KAIST , Daejeon, South Korea | Mar 2025 – Present |
| <ul style="list-style-type: none"> ▪ Associate Professor, Dept. of Industrial & Systems Engineering ▪ Associate Professor, The Kim Jaechul Graduate School of AI ▪ Associate Professor, Graduate School of Data Science | |
| KAIST , Daejeon, South Korea | Nov 2020 – Feb 2025 |
| <ul style="list-style-type: none"> ▪ Assistant Professor, Dept. of Industrial & Systems Engineering ▪ Assistant Professor, The Kim Jaechul Graduate School of AI ▪ Assistant Professor, Graduate School of Data Science | |
| University of Illinois at Urbana-Champaign , Urbana, IL, USA | Jan 2019 – Oct 2020 |
| <ul style="list-style-type: none"> ▪ Postdoctoral Reserach Fellow in Computer Science Department <ul style="list-style-type: none"> • Host: Prof. Jiawei Han • Topics: Multimodal data mining, Heterogeneous network representation learning | |
| Microsoft Research , Beijing, China | Sep 2017 – Dec 2017 |
| <ul style="list-style-type: none"> ▪ Research Intern <ul style="list-style-type: none"> • Mentor: Dr. Xing Xie • Project: Collaborative Translational Metric Learning | |
| NAVER , Seongnam, South Korea | Mar 2017 – Aug 2017 |
| <ul style="list-style-type: none"> ▪ Research Intern <ul style="list-style-type: none"> • Mentors: Dr. Jung-Tae Lee, and Dr. Min-Chul Yang • Project: Click-aware Purchase Prediction with Push at the Top | |

EDUCATION

- | | |
|--|----------|
| Pohang University of Science and Technology , Pohang, South Korea | Feb 2019 |
| <ul style="list-style-type: none"> ▪ Ph.D. in Computer Science and Engineering <ul style="list-style-type: none"> • Thesis: Recommendation Framework via Matrix Factorization and Translation • Topics: Multimodal data mining, User behavior modeling, Recommender system | |
| Sogang University , Seoul, South Korea | Feb 2014 |
| <ul style="list-style-type: none"> ▪ B.S. in Computer Science and Engineering | |

AWARDS & SCHOLARSHIPS

- | | |
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| Google Research Award | 2026 |
| <ul style="list-style-type: none"> ▪ Google | |
| Impact Research Award (Achievement Award) | 2025 |
| <ul style="list-style-type: none"> ▪ KAIST | |
| Best Academic Award | 2025 |

▪ Dept. of Industrial and Systems Engineering, KAIST	
Song-Am Distinguished Research Award	2024
▪ College of Engineering, KAIST	
Best Academic Award	2024
▪ Dept. of Industrial and Systems Engineering, KAIST	
Best Paper Award	2024
▪ KDD 2024 Workshop on Federated Learning for Data Mining and Graph Analytics (FedKDD), Barcelona, Spain	
Best Paper Award	2024
▪ KDD 2024 Workshop on Human-Interpretable AI, Barcelona, Spain	
Gold Prize, 30th Samsung Humantech Paper Award	2024
Technology Innovation Awards	2023
▪ College of Engineering, KAIST	
Top Reviewer of NeurIPS	2023
Best Paper Award	2023
▪ ICML 2023 Workshop on Computational Biology, Hawaii, USA	
KAIST Excellence in Teaching Award of Year 2022	2023
Best Teaching Award	2023
▪ Dept. of Industrial and Systems Engineering, KAIST	
ICDM Travel Award	2018
▪ International Conference on Data Mining, Singapore	
Award of excellence	2017
▪ Microsoft Research Asia Internship Program, Beijing, China	
▪ Awarded to the best intern	
Naver Ph.D Fellowship	2016
▪ Awarded to outstanding Ph.D students majoring in computer science in Korea	
RecSys Challenge 2015	2015
▪ Ranked 10 out of 850 teams from 49 countries (Top 1.1%)	
▪ Task: Click/Purchase Prediction	
Dean's List	Fall 2013
▪ Sogang University	

PUBLICATIONS

CONFERENCES

- [C86] Sink-Token-Aware Pruning for Fine-Grained Video Understanding in Efficient Video LLMs
Kibum Kim, Jiwan Kim, Kyle Min, Yueqi Wang, Jinyoung Moon, Julian McAuley, **Chanyoung Park**
European Conference on Computer Vision (**ECCV 2026**)
- [C85] Why and When Visual Token Pruning Fails? A Study on Relevant Visual Information Shift in MLLMs Decoding
Jiwan Kim, Kibum Kim, Wonjoong Kim, Byung-Kwan Lee, **Chanyoung Park**
European Conference on Computer Vision (**ECCV 2026**)
- [C84] PathCRF: Ball-Free Soccer Event Detection via Possession Path Inference from Player Trajectories
Hyunsung Kim, Kunhee Lee, Sangwoo Seo, Sang-Ki Ko, Jinsung Yoon, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2026**)
- [C83] Learning Probabilistic Compositional Representation of Crystalline Materials
Namkyeong Lee, Heewoong Noh, Gyoung S. Na, Jimeng Sun, Tianfan Fu, Marinka Zitnik, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2026 (AI for Science Track)**)
- [C82] EqGINO: Equivariant Geometry-Informed Fourier Neural Operators for 3D PDEs
Sungwon Kim, Juho Song, Seungmin Shin, Guimok Cho, Sangkook Kim, **Chanyoung Park**
International Conference on Machine Learning (**ICML 2026**)
- [C81] Beyond the Final Answer: Evaluating the Reasoning Trajectories of Tool-Augmented Agents
Wonjoong Kim, Sangwu Park, Yeonjun In, Sein Kim, Dongha Lee, **Chanyoung Park**
International Conference on Machine Learning (**ICML 2026**)

- [C80] Learning Adaptive Perturbation-Conditioned Contexts for Robust Transcriptional Response Prediction
Yinhua Piao, Hyomin Kim, Seonghwan Kim, Yunhak Oh, Junhyeok Jeon, Sang-Yeon Hwang, Jaechang Lim, Woo Youn Kim, **Chanyoung Park**, Sungsoo Ahn
International Conference on Machine Learning (**ICML 2026**)
- [C79] Time-PEFT: Temporal and Multichannel Complexity-Based Fine-Tuning for Time-Series Foundation Models
Jihye Na, Patara Trirat, **Chanyoung Park**, Jae-Gil Lee
International Conference on Machine Learning (**ICML 2026**)
- [C78] SelfJudge: Faster Speculative Decoding via Self-Supervised Judge Verification
Kanghoon Yoon, Minsub Kim, Sungjae Lee, Joonhyung Lee, Sunghyeon Woo, Yeonjun In, Se Jung Kwon, **Chanyoung Park**, Dongsoo Lee
International Conference on Machine Learning (**ICML 2026**)
- [C77] Reasoning Structure Matters for Safety Alignment of Reasoning Models
Yeonjun In, Wonjoong Kim, Sangwu Park, **Chanyoung Park**
The 64th Annual Meeting of the Association for Computational Linguistics (**ACL 2026**)
- [C76] IR-Agent: Expert-Inspired LLM Agents for Structure Elucidation from Infrared Spectra
Heewoong Noh, Namkyeong Lee, Gyoung S. Na, Kibum Kim, **Chanyoung Park**
The Fourteenth International Conference on Learning Representations (**ICLR 2026**)
- [C75] CompoDistill: Attention Distillation for Compositional Reasoning in Multimodal LLMs
Jiwan Kim, Kibum Kim, Sangwoo Seo, **Chanyoung Park**
The Fourteenth International Conference on Learning Representations (**ICLR 2026**)
- [C74] Token-Efficient Item Representation via Images for LLM Recommender Systems
Kibum Kim, Sein Kim, HongSeok Kang, Jiwan Kim, Heewoong Noh, Yeonjun In, Kanghoon Yoon, Jinoh Oh, Julian McAuley, **Chanyoung Park**
The Fourteenth International Conference on Learning Representations (**ICLR 2026**)
- [C73] PULSE: Socially-Aware User Representation Modeling Toward Parameter-Efficient Graph Collaborative Filtering
Doyun Choi*, Cheonwoo Lee*, Biniyam Aschalew Tolera, Taewook Ham, **Chanyoung Park**, Jaemin Yoo
The Web Conference (**WWW 2026**)
- [C72] RAG-Enhanced Collaborative LLM Agents for Drug Discovery
Namkyeong Lee, Edward De Brouwer, Ehsan Hajiramezanali, Tommaso Biancalani, **Chanyoung Park**, Gabriele Scalia
The 40th Annual AAAI Conference on Artificial Intelligence (**AAAI 2026**)
- [C71] 3D Interaction Geometric Pre-training for Molecular Relational Learning
Namkyeong Lee, Yunhak Oh, Heewoong Noh, Gyoung S. Na, Minkai Xu, Hanchen Wang, Tianfan Fu, **Chanyoung Park**
The Thirty-Ninth Annual Conference on Neural Information Processing Systems (**NeurIPS 2025 (Spotlight)**)
- [C70] Disentangling Hyperedges through the Lens of Category Theory
Yoonho Lee, Junseok Lee, Sangwoo Seo, Sungwon Kim, Yeongmin Kim, **Chanyoung Park**
The Thirty-Ninth Annual Conference on Neural Information Processing Systems (**NeurIPS 2025**)
- [C69] Training Robust Graph Neural Networks by Modeling Noise Dependencies
Yeonjun In, Kanghoon Yoon, Sukwon Yun, Kibum Kim, Sungchul Kim, **Chanyoung Park**
The Thirty-Ninth Annual Conference on Neural Information Processing Systems (**NeurIPS 2025**)
- [C68] Is Safety Standard Same for Everyone? User-Specific Safety Evaluation of Large Language Models
Yeonjun In, Wonjoong Kim, Kanghoon Yoon, Sungchul Kim, Mehrab Tanjim, Kibum Kim, **Chanyoung Park**
Conference on Empirical Methods in Natural Language Processing (**EMNLP 2025 Findings**)
- [C67] Disambiguation in Conversational Question Answering in the Era of LLM: A Survey
Md Mehrab Tanjim*, Yeonjun In*, Xiang Chen, Victor S. Burszty, Ryan A. Rossi, Sungchul Kim, Guang-Jie Ren, Vaishnavi Muppala, Shun Jiang, Yongsung Kim, **Chanyoung Park**
Conference on Empirical Methods in Natural Language Processing (**EMNLP 2025**)

- [C66] Target Circuit Matching in Large-Scale Netlists using GNN-Based Region Prediction
Sangwoo Seo, Jimin Seo, Yoonho Lee, Donghyeon Kim, Hyejin Shin, Banghyun Sung, **Chanyoung Park**
International Conference on Computer-Aided Design (**ICCAD 2025**)
- [C65] Lost in Sequence: Do Large Language Models Understand Sequential Recommendation?
Sein Kim*, HongSeok Kang*, Kibum Kim, Jiwan Kim, Donghyun Kim, Min-Chul Yang, Kwangjin Oh, Julian McAuley, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2025**)
- [C64] Oldie but Goodie: Re-illuminating Label Propagation on Graphs with Partially Observed Features
Sukwon Yun, Xin Liu, Yunhak Oh, Junseok Lee, Tianlong Chen, Tsuyoshi Murata, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2025**)
- [C63] Electron-Informed Coarse-Graining Molecular Representation Learning for Real-World Molecular Physics
Gyoung S. Na, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2025**)
- [C62] On Measuring Unnoticeability of Graph Adversarial Attacks: Observations, New Measure, and Applications
Hyeonsoo Jo*, Hyunjin Hwang*, Fanchen Bu, Soo Yong Lee, **Chanyoung Park**, Kijung Shin
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2025**)
- [C61] Thickness-aware E(3)-Equivariant 3D Mesh Neural Networks
Sungwon Kim, Namkyeong Lee, Yunyoung Doh, Seungmin Shin, Guimok Cho, Seung-Won Jeon, Sangkook Kim, **Chanyoung Park**
International Conference on Machine Learning (**ICML 2025**)
- [C60] Global Context-aware Representation Learning for Spatially Resolved Transcriptomics
Yunhak Oh*, Junseok Lee*, Yeongmin Kim, Sangwoo Seo, Namkyeong Lee, **Chanyoung Park**
International Conference on Machine Learning (**ICML 2025**)
- [C59] Disentangling and Generating Modalities for Recommendation in Missing Modality Scenarios
Jiwan Kim, Hongseok Kang, Sein Kim, Kibum Kim, **Chanyoung Park**
ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR 2025**)
- [C58] Dynamic Time-aware Continual User Representation Learning
Seungyeon Choi, Sein Kim, Hongseok Kang, Wonjoong Kim, **Chanyoung Park**
ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR 2025**)
- [C57] Subgraph Federated Learning for Local Generalization
Sungwon Kim, Yoonho Lee, Yunhak Oh, Namkyeong Lee, Sukwon Yun, Junseok Lee, Sein Kim, Carl Yang, **Chanyoung Park**
The Thirteenth International Conference on Learning Representations (**ICLR 2025 (Oral)**)
- [C56] Weakly Supervised Video Scene Graph Generation via Natural Language Supervision
Kibum Kim, Kanghoon Yoon, Yeonjun In, Jaehyeong Jeon, Jinyoung Moon, Donghyun Kim, **Chanyoung Park**
The Thirteenth International Conference on Learning Representations (**ICLR 2025**)
- [C55] Self-Supervised Diffusion Models for Electron-Aware Molecular Representation Learning
Gyoung S. Na, **Chanyoung Park**
The Thirteenth International Conference on Learning Representations (**ICLR 2025**)
- [C54] Diversify-verify-adapt: Efficient and Robust Retrieval-Augmented Ambiguous Question Answering
Yeonjun In, Sungchul Kim, Ryan A. Rossi, Mehrab Tanjim, Tong Yu, Ritwik Sinha, **Chanyoung Park**
Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (**NAACL 2025**)
- [C53] SIMPLOT: Enhancing Chart Question Answering by Distilling Essentials
Wonjoong Kim*, Sangwu Park*, Yeonjun In, Seokwon Han, **Chanyoung Park**
Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (**NAACL 2025 Findings**)

- [C52] RA-SGG: Retrieval-Augmented Scene Graph Generation Framework via Multi-Prototype Learning
Kanghoon Yoon, Kibum Kim, Jaehyeong Jeon, Yeonjun In, Donghyun Kim, **Chanyoung Park**
Thirty-Ninth AAAI Conference on Artificial Intelligence (**AAAI 2025**)
- [C51] Revisiting Fake News Detection: Towards Temporality-aware Evaluation by Leveraging Engagement Earliness
Junghoon Kim*, Junmo Lee*, Yeonjun In, Kanghoon Yoon, **Chanyoung Park**
The 18th ACM International Conference on Web Search and Data Mining (**WSDM 2025 (Oral)**)
- [C50] Retrieval-Retro: Retrieval-based Inorganic Retrosynthesis with Expert Knowledge
Heewoong Noh, Namkyeong Lee, Gyoung S. Na, **Chanyoung Park**
Thirty-eighth Conference on Neural Information Processing Systems (**NeurIPS 2024**)
- [C49] Debaised Graph Poisoning Attack via Contrastive Surrogate Objective
Kanghoon Yoon, Yeonjun In, Namkyeong Lee, Kibum Kim, **Chanyoung Park**
ACM International Conference on Information and Knowledge Management (**CIKM 2024**)
- [C48] Vision Language Model is NOT All You Need: Augmentation Strategies for Molecule Language Models
Namkyeong Lee, Siddhartha Laghuvarapu, **Chanyoung Park**, Jimeng Sun
ACM International Conference on Information and Knowledge Management (**CIKM 2024**)
- [C47] Semantic Diversity-aware Prototype-based Learning for Unbiased Scene Graph Generation
Jaehyeong Jeon, Kibum Kim, Kanghoon Yoon, **Chanyoung Park**
European Conference on Computer Vision (**ECCV 2024**)
- [C46] Multiplexed Immunofluorescence Image Analysis through an Efficient Multiplex Network
Sukwon Yun, Jie Peng, Alexandro Trevino, **Chanyoung Park**, Tianlong Chen
European Conference on Computer Vision (**ECCV 2024**)
- [C45] Large Language Models meet Collaborative Filtering: An Efficient All-round LLM-based Recommender System
Sein Kim*, Hongseok Kang*, Seungyeon Choi, Donghyun Kim, Minchul Yang, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2024**)
- [C44] Self-Explainable Temporal Graph Networks based on Graph Information Bottleneck
Sangwoo Seo, Sungwon Kim, Jihyeong Jung, Yoonho Lee, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2024**)
- [C43] LLM4SGG: Large Language Models for Weakly Supervised Scene Graph Generation
Kibum Kim, Kanghoon Yoon, Jaehyeong Jeon, Yeonjun In, Jinyoung Moon, Donghyun Kim, **Chanyoung Park**
IEEE/CVF Computer Vision and Pattern Recognition Conference (**CVPR 2024**)
- [C42] Self-guided Robust Graph Structure Refinement
Yeonjun In, Kanghoon Yoon, Kibum Kim, Kijung Shin, **Chanyoung Park**
The Web Conference 2024 (**WWW 2024 (Oral)**)
- [C41] DSLR: Diversity Enhancement and Structure Learning for Rehearsal-based Graph Continual Learning
Seungyeon Choi*, Wonjoong Kim*, Sungwon Kim, Yeonjun In, Sein Kim, **Chanyoung Park**
The Web Conference 2024 (**WWW 2024 (Oral)**)
- [C40] Adaptive Self-training Framework for Fine-grained Scene Graph Generation
Kibum Kim*, Kanghoon Yoon*, Yeonjun In, Jinyoung Moon, Donghyun Kim, **Chanyoung Park**
Twelfth International Conference on Learning Representations (**ICLR 2024**)
- [C39] Sterling: Synergistic Representation Learning on Bipartite Graphs
Baoyu Jing, Yuchen Yan, Kaize Ding, **Chanyoung Park**, Yada Zhu, Huan Liu, Hanghang Tong
Thirty-Eighth AAAI Conference on Artificial Intelligence (**AAAI 2024 (Oral)**)
- [C38] Interpretable Prototype-based Graph Information Bottleneck
Sangwoo Seo, Sungwon Kim, **Chanyoung Park**
Thirty-seventh Conference on Neural Information Processing Systems (**NeurIPS 2023**) (**Gold Prize, 30th Samsung Humantech Paper Award**)

- [C37] Density of States Prediction of Crystalline Materials via Prompt-guided Multi-Modal Transformer
Namkyeong Lee*, Heewoong Noh*, Sungwon Kim, Dongmin Hyun, Gyoung S. Na, **Chanyoung Park**
Thirty-seventh Conference on Neural Information Processing Systems (**NeurIPS 2023**)
- [C36] MUSE: Music Recommender System with Shuffle Play Recommendation Enhancement
Yunhak Oh*, Sukwon Yun*, Dongmin Hyun, Sein Kim, **Chanyoung Park**
ACM International Conference on Information and Knowledge Management (**CIKM 2023**)
- [C35] Class Label-aware Graph Anomaly Detection
Junghoon Kim, Yeonjun In, Kanghoon Yoon, Junmo Lee, **Chanyoung Park**
ACM International Conference on Information and Knowledge Management (**CIKM 2023**) (Short Paper)
- [C34] Structural Mixup for Graph Neural Networks
Junghun Kim*, Sukwon Yun*, **Chanyoung Park**
ACM International Conference on Information and Knowledge Management (**CIKM 2023**) (Short Paper)
- [C33] Similarity Preserving Adversarial Graph Contrastive Learning
Yeonjun In*, Kanghoon Yoon*, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2023**)
- [C32] Shift-Robust Molecular Relational Learning with Causal Substructure
Namkyeong Lee, Kanghoon Yoon, Gyoung S. Na, Sein Kim, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2023**)
- [C31] Task-Equivariant Graph Few-shot Learning
Sungwon Kim, Junseok Lee, Namkyeong Lee, Wonjoong Kim, Seungyoon Choi, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2023**)
- [C30] Task Relation-aware Continual User Representation Learning
Sein Kim, Namkyeong Lee, Donghyun Kim, Min-Chul Yang, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2023**)
- [C29] Conditional Graph Information Bottleneck for Molecular Relational Learning
Namkyeong Lee, Dongmin Hyun, Gyoung S. Na, Sungwon Kim, Junseok Lee, **Chanyoung Park**
International Conference on Machine Learning (**ICML 2023**)
- [C28] MELT: Mutual Enhancement of Long-Tailed User and Item for Sequential Recommendation
Kibum Kim, Dongmin Hyun, Sukwon Yun, **Chanyoung Park**
ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR 2023**)
- [C27] Unbiased Heterogeneous Scene Graph Generation with Relation-aware Message Passing Neural Network
Kanghoon Yoon*, Kibum Kim*, Jinyoung Moon, **Chanyoung Park**
Thirty-Seventh AAAI Conference on Artificial Intelligence (**AAAI 2023**)
- [C26] Heterogeneous Graph Learning for Multi-modal Medical Data Analysis
Sein Kim, Namkyeong Lee, Junseok Lee, Dongmin Hyun, **Chanyoung Park**
Thirty-Seventh AAAI Conference on Artificial Intelligence (**AAAI 2023 (Oral)**)
- [C25] Generating Multiple-Length Summaries via Reinforcement Learning for Unsupervised Sentence Summarization
Dongmin Hyun, Xiting Wang, **Chanyoung Park**, Xing Xie, Hwanjo Yu
Conference on Empirical Methods in Natural Language Processing (**EMNLP 2022 Findings**)
- [C24] Set2Box: Similarity Preserving Representation Learning for Sets
Geon Lee, **Chanyoung Park**, Kijung Shin
The IEEE International Conference on Data Mining (**ICDM 2022**)
- [C23] LTE4G: Long-Tail Experts for Graph Neural Networks
Sukwon Yun, Kibum Kim, Kanghoon Yoon, **Chanyoung Park**
ACM International Conference on Information and Knowledge (**CIKM 2022**)
- [C22] Relational Self-Supervised Learning on Graphs
Namkyeong Lee, Dongmin Hyun, Junseok Lee, **Chanyoung Park**
ACM International Conference on Information and Knowledge (**CIKM 2022**)

- [C21] Beyond Learning from Next Item: Sequential Recommendation via Personalized Interest Sustainability
Dongmin Hyun, **Chanyoung Park**, Junsu Cho, Hwanjo Yu
ACM International Conference on Information and Knowledge (**CIKM 2022**)
- [C20] Nonlinearity Encoding for Extrapolation of Neural Networks
Gyoung S. Na, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2022**)
- [C19] GraFN: Semi-Supervised Node Classification on Graph with Few Labels via Non-Parametric Distribution Assignment
Junseok Lee, Yunhak Oh, Yeonjun In, Namkyeong Lee, Dongmin Hyun, **Chanyoung Park**
ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR 2022**) (Short Paper)
- [C18] AHP: Learning to Negative Sample for Hyperedge Prediction
Hyunjin Hwang, Seungwoo Lee, **Chanyoung Park**, Kijung Shin
ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR 2022**) (Short Paper)
- [C17] Augmentation-Free Self-Supervised Learning on Graphs
Namkyeong Lee, Junseok Lee, **Chanyoung Park**
Thirty-Sixth AAAI Conference on Artificial Intelligence (**AAAI 2022**)
- [C16] Learnable Structural Semantic Readout for Graph Classification
Dongha Lee, Su Kim, Seonghyeon Lee, **Chanyoung Park**, Hwanjo Yu
IEEE International Conference on Data Mining (**ICDM 2021**) (Short Paper)
- [C15] Bootstrapping User and Item Representations for One-Class Collaborative Filtering
Dongha Lee, Seongku Kang, Hyunjun Ju, **Chanyoung Park**, Hwanjo Yu
ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR 2021**)
- [C14] HDMI: High-order Deep Multiplex Infomax
Baoyu Jing, **Chanyoung Park**, Hanghang Tong
The Web Conference (**WWW 2021**)
- [C13] Interest Sustainability-Aware Recommender System
Dongmin Hyun, Junsu Cho, **Chanyoung Park**, Hwanjo Yu
IEEE International Conference on Data Mining (**ICDM 2020**)
- [C12] Unsupervised Differentiable Multi-aspect Network Embedding
Chanyoung Park, Carl Yang, Qi Zhu, Donghyun Kim, Hwanjo Yu and Jiawei Han
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2020**)
- [C11] Unsupervised Attributed Multiplex Network Embedding
Chanyoung Park, Donghyun Kim, Jiawei Han, Hwanjo Yu
Thirty-Fourth AAAI Conference on Artificial Intelligence (**AAAI 2020**)
- [C10] Task-Guided Pair Embedding in Heterogeneous Network
Chanyoung Park, Donghyun Kim, Qi Zhu, Jiawei Han, Hwanjo Yu
ACM International Conference on Information and Knowledge (**CIKM 2019**)
- [C9] BHIN2vec: Balancing the Type of Relation in Heterogeneous Information Network
Seonghyun Lee, **Chanyoung Park**, Hwanjo Yu
ACM International Conference on Information and Knowledge (**CIKM 2019**)
- [C8] Action Space Learning for Heterogeneous User Behavior Prediction
Dongha Lee, **Chanyoung Park**, Hyunjun Ju, Junyoung Hwang, Hwanjo Yu
International Joint Conference on Artificial Intelligence (**IJCAI 2019**)
- [C7] Sequential and Diverse Recommendation with Long Tail
Yejin Kim, Kwangseob Kim, **Chanyoung Park**, Hwanjo Yu
International Joint Conference on Artificial Intelligence (**IJCAI 2019**)
- [C6] Collaborative Translational Metric Learning
Chanyoung Park, Donghyun Kim, Xing Xie, Hwanjo Yu
IEEE International Conference on Data Mining (**ICDM 2018**)

- [C5] Review Sentiment-Guided Scalable Deep Recommender System
Dongmin Hyun, **Chanyoung Park**, Min-Chul Yang, Ilhyeon Song, Jung-Tae Lee, Hwanjo Yu
ACM SIGIR conference on Research and Development in Information Retrieval (**SIGIR 2018**)
(Short Paper)
- [C4] Do “Also-Viewed” Products Help User Rating Prediction?
Chanyoung Park, Donghyun Kim, Jinoh Oh, Hwanjo Yu
International World-Wide Web Conference (**WWW 2017**)
- [C3] Convolutional Matrix Factorization for Document Context-Aware Recommendation
Donghyun Kim, **Chanyoung Park**, Jinoh Oh, Hwanjo Yu
ACM international conference on Recommender System (**RecSys 2016**)
- [C2] TRecSo: Enhancing Top-k Recommendation With Social Information
Chanyoung Park, Donghyun Kim, Jinoh Oh, Hwanjo Yu
International World-Wide Web Conference (**WWW 2016**) (Poster)
- [C1] Predicting User Purchase in E-commerce by Comprehensive Feature Engineering and Decision
Boundary Focused Under-Sampling
Chanyoung Park, Donghyun Kim, Jinoh Oh, Hwanjo Yu
ACM international conference on Recommender System Challenge (**ACM RecSysChallenge 2015**,
10th place, top 1.1%)

JOURNALS

- [J14] A New DAWN for Fake News Detection: Leveraging Engagement Earliness for Temporality-aware
Evaluation
Junghoon Kim*, Junmo Lee*, Yeonjun In, Kanghoon Yoon, **Chanyoung Park**
ACM Transactions on Intelligent Systems and Technology (2026)
- [J13] Physics-Embedded Graph Neural Operator for Interaction-Controlled Colloidal Aggregation
Yongjoon Choe*, Sungwon Kim*, Susan E. Burns, **Chanyoung Park**
Water Research (2026)
- [J12] Single-cell RNA Sequencing Data Imputation Using Bi-level Feature Propagation
Junseok Lee*, Sukwon Yun*, Yeongmin Kim, Tianlong Chen, Manolis Kellis, **Chanyoung Park**
Briefings in Bioinformatics (2024)
- [J11] Deep Single-cell RNA-seq Data Clustering with Graph Prototypical Contrastive Learning
Junseok Lee, Sungwon Kim, Dongmin Hyun, Namkyeong Lee, Yejin Kim, **Chanyoung Park**
Bioinformatics (2023)
- [J10] Self-Supervised Graph Representation Learning via Positive Mining
Namkyeong Lee, Junseok Lee, **Chanyoung Park**
Information Sciences (2022.9)
- [J9] Learning to Utilize Auxiliary Reviews for Recommendation
Dongmin Hyun, **Chanyoung Park**, Junsu Cho, Hwanjo Yu
Information Sciences (2021.2)
- [J8] Deep Multiplex Graph Infomax: Attentive Multiplex Network Embedding using Global Information
Chanyoung Park, Jiawei Han, Hwanjo Yu
Knowledge-Based Systems (2020.6)
- [J7] Improved prediction of miRNA–disease associations based on matrix completion with network
regularization
Jihwan Ha, Chihyun Park, **Chanyoung Park**, Sanghyun Park
Cells (2020.4) (SCIE)
- [J6] Click-aware Purchase Prediction with Push at the Top
Chanyoung Park, Donghyun Kim, Min-Chul Yang, Jung-Tae Lee, Hwanjo Yu
Information Sciences (2020.6)
- [J5] Inferring miRNA-disease Interactions using Probabilistic Matrix Factorization
Jihwan Ha, Chihyun Park, **Chanyoung Park**, Sanghyun Park
Journal of Biomedical Informatics (2020.2)
- [J4] An Encoder-Decoder Switch Network for Purchase Prediction
Chanyoung Park, Donghyun Kim, Hwanjo Yu
Knowledge-Based Systems (2019.12)

- [J3] Target-aware Convolutional Neural Network for Target-level Sentiment Analysis
Dongmin Hyun, **Chanyoung Park**, Min-Chul Yang, Ilhyeon Song, Jung-Tae Lee, Hwanjo Yu
Information Sciences (2019.7)
- [J2] Deep Hybrid Recommender Systems via Exploiting Document Context and Statistics of Items
Donghyun Kim, **Chanyoung Park**, Jinoh Oh, Hwanjo Yu
Information Sciences (2017.11)
- [J1] Improving top-K recommendation with truster and trustee relationship in user trust network
Chanyoung Park, Donghyun Kim, Jinoh Oh, Hwanjo Yu
Information Sciences (2016.12)

WORKSHOPS

- [W16] EqGINO: Equivariant Geometry-Informed Fourier Neural Operators for 3D PDEs
Sungwon Kim, Juho Song, Seungmin Shin, Guimok Cho, Sangkook Kim, **Chanyoung Park**
ICLR 2026 Workshop on AI and Partial Differential Equations (Oral) (ICLR 2026 Workshop)
- [W15] Self-EvolveRec: Self-Evolving Recommender Systems with LLM-based Directional Feedback
Sein Kim, Sangwu Park, HongSeok Kang, Wonjoong Kim, Jimin Seo, Yeonjun In, Kanghoon Yoon,
Chanyoung Park
ICLR 2026 Workshop on AI with Recursive Self-Improvement (RSI 2026) (ICLR 2026 Workshop)
- [W14] MSP-LLM: A Unified Large Language Model Framework for Complete Material Synthesis Planning
Heewoong Noh, Gyoung S. Na, Namkyeong Lee, **Chanyoung Park**
AI4Mat: AI for Accelerated Materials Design (ICLR 2026 Workshop)
- [W13] Synergistic Multi-Task Learning for Electronic Density of States Prediction
Kunmin Jang, Dongik Park, Jaewon Bae, **Chanyoung Park**
AI4Mat: AI for Accelerated Materials Design (ICLR 2026 Workshop)
- [W12] Learning k-Resolved Electronic Structure via Soft Energy Occupancy Prediction
Dongik Park, Kunmin Jang, Jaewon Bae, Dongin Kim, **Chanyoung Park**
AI4Mat: AI for Accelerated Materials Design (ICLR 2026 Workshop)
- [W11] Learning Adaptive Perturbation-Conditioned Contexts for Robust Transcriptional Response Prediction
Yinhua Piao, Hyomin Kim, Seonghwan Kim, Yunhak Oh, Junhyeok Jeon, Sang-Yeon Hwang, Jaechang Lim, Woo Youn Kim, **Chanyoung Park**, Sungsoo Ahn
ICLR 2026 Workshop on Generative and Experimental Perspectives for Biomolecular Design (ICLR 2026 Workshop)
- [W10] RAG-Enhanced Collaborative LLM Agents for Drug Discovery
Namkyeong Lee, Edward De Brouwer, Ehsan Hajiramezanali, **Chanyoung Park**, Gabriele Scalia
MLGenX Workshop at ICLR 2025 (MLGenX 2025 Spotlight)
- [W9] 3D Interaction Geometric Pre-training for Molecular Relational Learning
Namkyeong Lee, Yunhak Oh, Heewoong Noh, Gyoung S. Na, Tianfan Fu, **Chanyoung Park**
NeurIPS 2024 Workshop on AI for New Drug Modalities (AIDrugX)
- [W8] Subgraph Federated Learning for Local Generalization
Sungwon Kim, Yoonho Lee, Yunhak Oh, Namkyeong Lee, Sukwon Yun, Junseok Lee, Sein Kim, Carl Yang, **Chanyoung Park**
KDD 2024 Workshop on Human-Interpretable AI (Oral) (Best Paper Award)
- [W7] Interpretable Graph Model with Prototype-Based Graph Information Bottleneck
Sangwoo Seo, Sungwon Kim, **Chanyoung Park**
KDD 2024 Workshop on Human-Interpretable AI (Oral) (Best Paper Award)
- [W6] Stoichiometry Representation Learning with Polymorphic Crystal Structures
Namkyeong Lee, Heewoong Noh, Gyoung S. Na, Tianfan Fu, Jimeng Sun, **Chanyoung Park**
NeurIPS 2023 Workshop on AI for Scientific Discovery: From Theory to Practice (AI4Science)
- [W5] Electron-Derived Molecular Representation Learning for Real-World Molecular Physics
Gyoung S. Na, **Chanyoung Park**
NeurIPS 2023 Workshop on AI for Scientific Discovery: From Theory to Practice (AI4Science)

- [W4] Deep single-cell RNA-seq data clustering with graph prototypical contrastive learning
Junseok Lee, Sungwon Kim, Dongmin Hyun, Namkyeong Lee, Yejin Kim, **Chanyoung Park**
ICML 2023 Workshop on Computational Biology
- [W3] Denoising single-cell RNA-seq data using Feature Propagation
Sukwon Yun, Junseok Lee, **Chanyoung Park**
ICML 2023 Workshop on Computational Biology (Oral) (Best Paper Award)
- [W2] Predicting Density of States via Multi-modal Transformer
Namkyeong Lee, Heewoong Noh, Sungwon Kim, Dongmin Hyun, Gyoung S. Na, **Chanyoung Park**
ICLR 2023 Workshop on Machine Learning for Materials (ML4Materials)
- [W1] Shift-Robust Node Classification via Graph Adversarial Clustering
Qi Zhu, Chao Zhang, **Chanyoung Park**, Carl Yang, Jiawei Han
NeurIPS 2022 GLFrontiers Workshop

PATENT

- [P1] Method for context-aware recommendation by considering contextual information of document and apparatus for the same
Donghyun Kim, **Chanyoung Park**, Jinoh Oh, Hwanjo Yu
No. 10-18771610000, issued July 2018.

PREPRINT

- [A1] Adaptive Graph Rewiring to Mitigate Over-Squashing in Mesh-Based GNNs for Fluid Dynamics Simulations
Sangwoo Seo, Hyunsung Kim, Jiwan Kim, **Chanyoung Park**

TEACHING

Instructor

- IE343: Statistical Machine Learning, KAIST 2021S, 2022S, 2023S, 2024S, 2026S
- DS535 (KSE801): Recommender System and Machine Learning on Graphs, KAIST 2021F, 2022F, 2023F, 2024F
- CoE202: Basics of Artificial Intelligence, KAIST 2021F
- DS503: Machine Learning for Data Science (Deep Learning), KAIST 2022S, 2023S, 2024S

Teaching Assistant

- CSED352: Data Communications, POSTECH 2014F
- CSED312: Operating System, POSTECH 2016F

PROFESSIONAL SERVICES

AREA CHAIR

- The International Conference on Machine Learning (ICML) [2025-present]
- The International Conference on Learning Representations (ICLR) [2025-present]
- ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) [2025-present]
- Conference on Neural Information Processing Systems (NeurIPS) [2024-present]
- ICML 2024 AI4Science Workshop [2024-present]
- NeurIPS Workshop (AI for Science: from Theory to Practice) [2023-present]

PROGRAM COMMITTEE/REVIEWER

- ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) [2022-present]
- The Web Conference (WWW) [2020-present]
- AAAI Conference on Artificial Intelligence (AAAI), [2021-present]
- Conference on Neural Information Processing Systems (NeurIPS) [2022-present]
- International Conference on Machine Learning (ICML) [2023-present]
- The International Conference on Learning Representations (ICLR), [2022-present]
- ACM International Conference on Research and Development in Information Retrieval (SIGIR) [2023-present]
- ACM International Web Search and Data Mining Conference (WSDM) [2023-present]
- International Joint Conference on Artificial Intelligence (IJCAI) [2023-present]
- ACM International Conference on Information and Knowledge Management (CIKM) [2023-present]
- SIAM International Conference on Data Mining (SDM) [2023-present]
- Learning on Graphs Conference (LoG) [2022-present]
- International ACM SIGIR Conference on Information Retrieval in the Asia Pacific (SIGIR-AP) [2023-present]
- The Web Conference (WWW) Poster Track, 2020

- ACM International Conference on Information and Knowledge Management (CIKM) Short Paper Track, [2020-present]
- IEEE International Conference on Big Data (BigData) [2020-2023]
- ECML-PKDD Research Track [2023-present]
- ECML-PKDD Applied Data Science Track and Demo Track [2020-2021]
- International Conference on Database Systems for Advanced Applications (DASFAA), 2021
- International Conference on Big Data and Smart Computing (BigComp) 2023
- International Conference on Internet and Web Applications and Services (ICIW) [2017-2018]
- Graph Learning @ TheWebConf, 2022
- International Joint Conference on Artificial Intelligence (IJCAI) - Special Track on AI for Good, 2022
- NeurIPS Workshop (New Frontiers of AI for Drug Discovery and Development) [2023-present]

EVENT ORGANIZATIONS

- Organizer, KDD'26 Workshop on SciSoc Agents & LLMs [2026]
- General Chair, The ACM International Conference on Information and Knowledge Management (CIKM) [2025]
- Proceedings Chair, The ACM International Conference on Information and Knowledge Management (CIKM) [2023]

JOURNAL EDITORSHIP

- Action Editor in Transactions on Machine Learning Research ([TMLR](#))
- Frontiers in Artificial Intelligence ([Graph representation learning: Methods and Applications](#))
- Frontiers in Big Data ([Reviews in Big Data Networks: 2022](#))